# Naveed Salman

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**Right to Work in the UK: British Citizen (no sponsorship required)**

**Portfolio: https://nav-101.github.io/Nav-101/**

## Electronics and Systems Engineer

**Mixed-Signal and High-Speed Design | Embedded Systems | Wireless & Signal Processing | Systems Engineering**

Experienced electronics and embedded systems engineer with over a decade of experience spanning **hardware design, embedded systems, wireless communications, signal processing, and system-level engineering**. Proven ability to take complex systems from concept through design, implementation, and validation across **industrial, research, and applied engineering environments**. Strong hands-on background complemented by deep analytical expertise in estimation, tracking, and data fusion. Comfortable operating at the intersection of hardware, firmware, and algorithms.

## CORE TECHNICAL EXPERTISE:

- **Electronic & PCB Design**: Mixed-signal circuit design, precision analogue front-ends, power management (DC-DC, LDOs), low-noise design, EMC-aware layout, multilayer PCBs
- **Embedded Systems**: ARM Cortex-M, ESP32/ESP8266, real-time embedded firmware, hardware bring-up, debugging
- **Interfaces & Protocols**: Memory – SDRAM, NOR FLASH, SPI, $I^2C$, UART, USB, CAN, Ethernet, Analogue, Audio codec.
- **Wireless & Signal Processing**: Localisation, tracking, Bayesian filtering, wireless channels, OFDM, Wi-Fi, sensor networks
- **Data & Algorithms**: Python (NumPy, Pandas, Matplotlib), MATLAB, estimation & optimisation, data fusion
- **Systems Engineering**: End-to-end system design, cross-disciplinary integration, verification and validation

## PROFESSIONAL EXPERIENCE

**Coventry University,** *Coventry UK***.** 01/2022 - PRESENT

**Lecturer / Deputy Course Director - Electrical & Electronic Engineering.**

***Engineering-focused role combining system design, embedded hardware development, and industry-aligned delivery.***

- Designed and maintained **embedded hardware platforms** used in undergraduate, postgraduate, and industrial training environments

- Delivered **automotive networking and embedded systems training** to engineers as part of **JLR** electrification upskilling programmes
- Led instruction in **PCB design, embedded systems, intelligent wireless systems, and digital electronics** with strong practical emphasis
- Supervised and technically guided multiple hardware-centric student projects involving MCU-based systems, sensors, and communications
- Acted as **Deputy Course Director**, contributing to curriculum design aligned with industry expectations and engineering practice

**University of Leeds,** *Leeds, UK.* 01/2018 - 01/2022

**Research Fellow - School of Civil Engineering.**

***Applied engineering role focused on real-world sensing systems and data-driven system optimisation.***

- Designed and deployed **wireless air-quality sensing systems** for the EPSRC HECOIRA project
- Developed **embedded sensor nodes** and end-to-end data pipelines operating in live hospital environments
- Implemented **data assimilation and estimation algorithms** to improve CFD-based system predictions
- Worked closely with multidisciplinary engineering teams, translating theoretical models into deployable systems

**Sheffield Hallam University**, *Sheffield, UK.* 06/2016 - 01/2018

**Researcher - National Centre of Excellence for Food Engineering.**

***Industry-facing engineering and data analysis role.***

- Analysed large-scale industrial process data for an **Innovate UK project with Nestlé**
- Identified correlations between sensor data and product quality metrics
- Conducted on-site engineering visits to industrial facilities and presented technical findings to stakeholders

**University of Sheffield,** *Sheffield, UK* 06/ 2014 - 06/2016

**Research Associate - Department of Automatic Control and Systems Engineering.**

***Advanced systems and algorithm development role.***

- Core contributor to the EPSRC project **"Bayesian Tracking and Reasoning over Time (BTaRoT)"**
- Designed and implemented **particle filters, MCMC methods, and tracking algorithms**
- Produced validated results adopted in peer-reviewed publications

**Etisalat British Telecom Innovation Centre,** *Abu Dhabi, UAE* 2013

### Research Intern

- Developed Wi-Fi-based localisation and ranging techniques using MUSIC and OFDM signal processing
- Conducted extensive research on signal processing and wireless communication methods

## EDUCATION

**University of Leeds,** *Leeds, UK* 2009-2013
**Ph.D. in Electronic and Electrical Engineering,** Optimized Localisation in Wireless Sensor Networks.

**University of Leeds,** *Leeds, UK* 2008-2009
**M.Sc. in Electronic and Electrical Engineering,** Modern digital and wireless communications

**University of Engineering and Technology Peshawar,** *Peshawar, Pakistan* 2004-2007
**B.Sc. in Electronic and Electrical Engineering,** Wireless communications

**Coventry University,** *Coventry UK.* 2023
**Postgraduate Certificate in Academic Practice in HE (PgCAPHE),**

## SELECTED PUBLICATIONS

- N. Salman, A. Khan, A.H. Kemp, C.J. Noakes. Indoor Temperature Forecast based on the Lattice Boltzmann method and Data Assimilation. Building and Environment. 2022
- N. Salman, M.W. Khan, M. Lim, A. Khan, A.H. Kemp, C.J. Noakes. Use of Multiple Low-Cost Carbon Dioxide Sensors to Measure Exhaled Breath Distribution with Face Mask Type and Wearing Behaviour. Sensors, 21(17). 2021
- N. Salman, A.H. Kemp, A. Khan, C. Noakes. Real-Time Wireless Sensor Network (WSN)-Based Indoor Air Quality Monitoring System. 5th IFAC Symposium on Telematics, Chengdu, China. 2019

Complete list of publication on my Google Scholar page: Google Scholar link